PARINVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

Department of CSE(AI&ML)

AIML CLUB

PRESENTS

CNN: Powering Visual Intelligence in AI

GAURAV KSHIRSAGAR
(MEMBER)

VAIBHAV BURA
(MEMBER)

Key Takeaways :

- Foundations of Convolutional Neural Networks (CNN)
- CNN architecture explained visually & intuitively
- Hands-on demos with image data
- Real-world application

JOIN US!

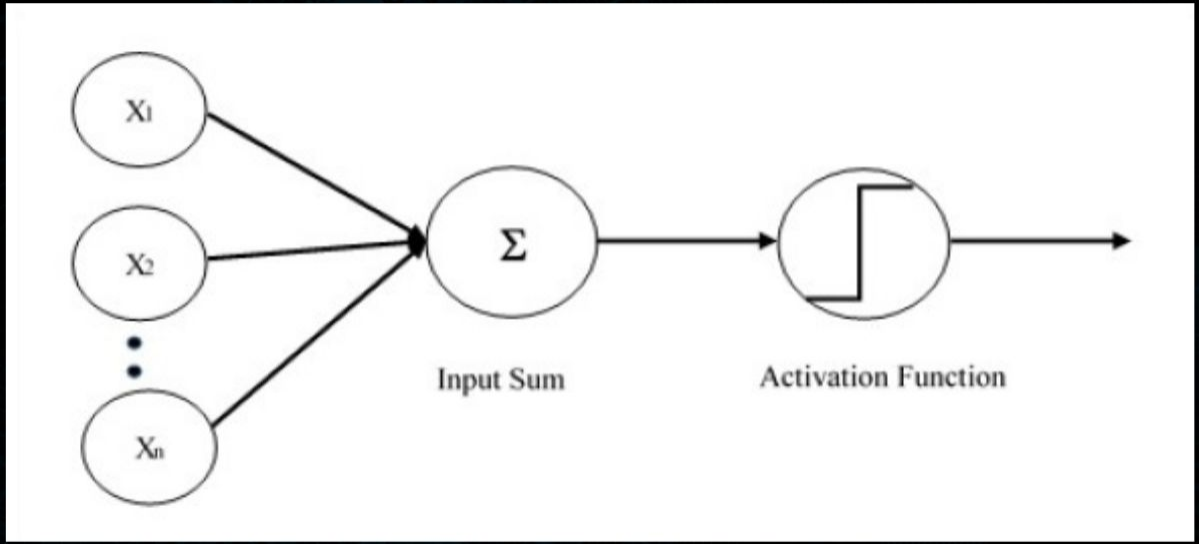
Date: **17th February**

Time: **08:30 am onwards**

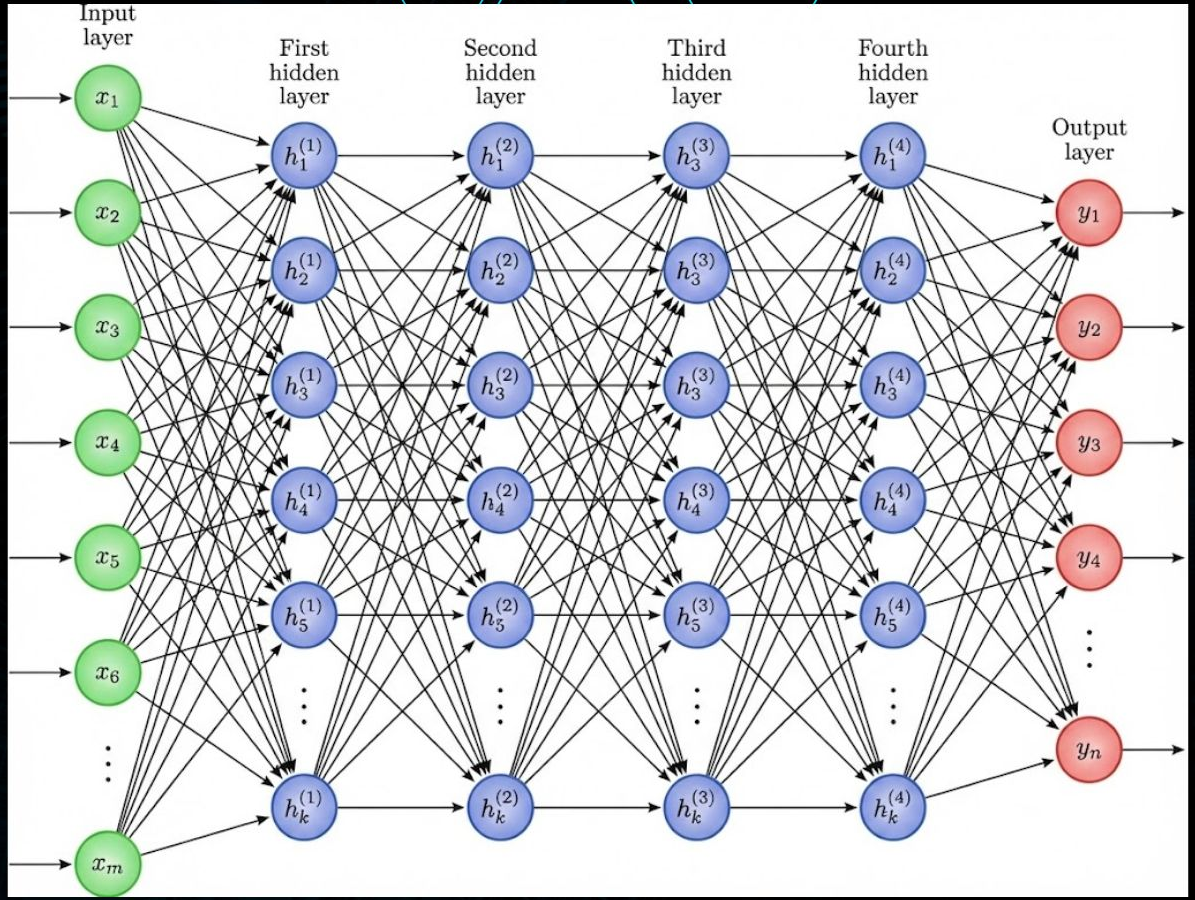
Venue: **405**

Pixels to Patterns

CNN: Powering Visual Intelligence in AI



Single Layer Perceptron – How a Neuron Makes a Decision

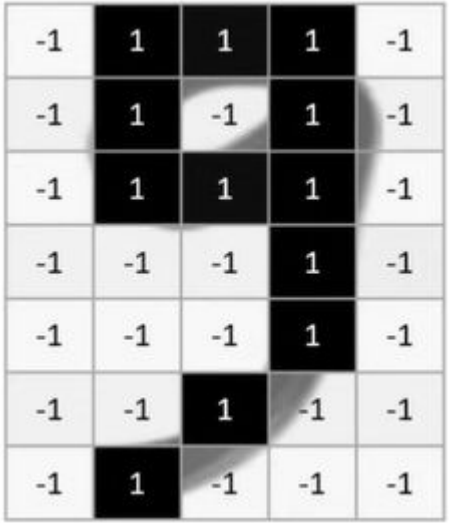


Multilayer Perceptron (MLP) – Learning in Multiple Stages

Humans recognize images instantly.

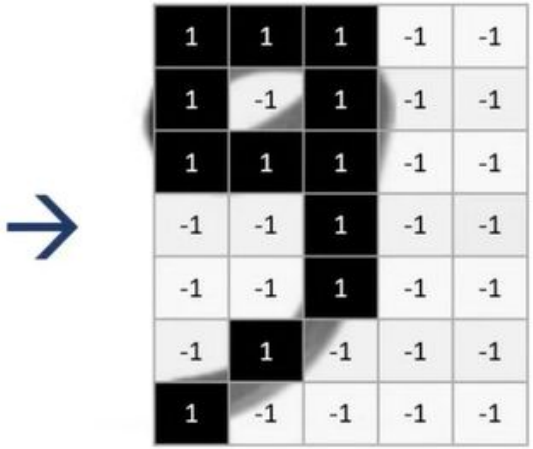
For humans, this is 9. For computers, it’s numbers

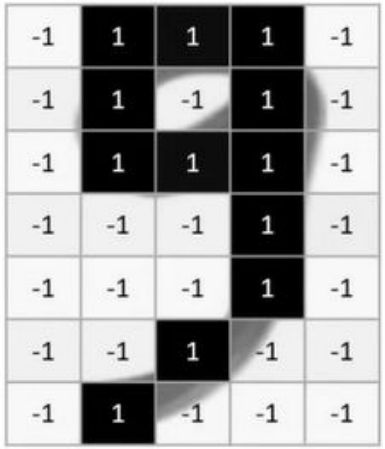
-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1



-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

Location shifted

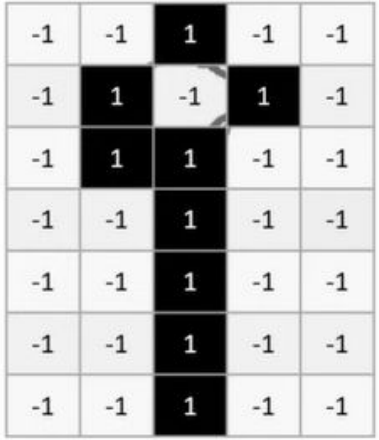




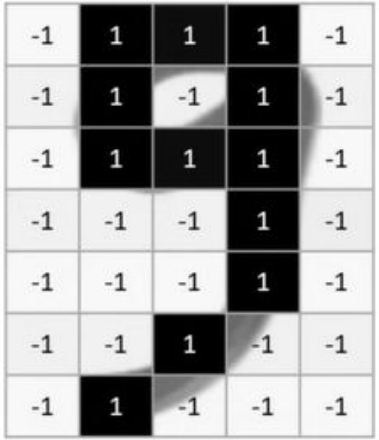
When the position of an object changes, pixel values change - ANN fails to recognize it.

Variation 1

-1	-1	1	-1	-1
-1	1	-1	1	-1
-1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1



-1	-1	1	-1	-1
-1	1	-1	1	-1
-1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1



-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

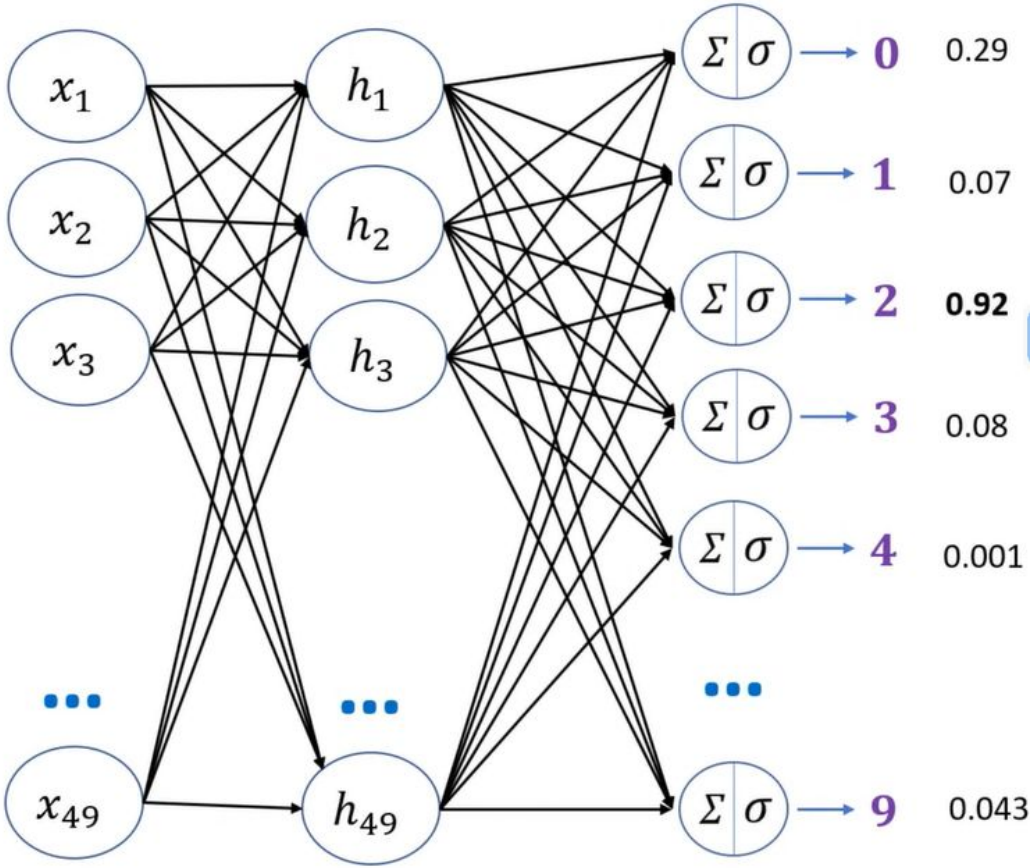
Handwriting varies; machines need feature understanding, not raw numbers.

SECTION 2: WHY ANN FAILS

0	0	0	0	0	0	0
0	87	240	210	24	0	0
0	13	0	101	195	0	0
0	35	167	99	210	0	0
0	145	230	240	201	189	140
0	0	102	67	17	13	0
0	0	0	0	0	0	0

7 by 7 grid

0
0
0
0
0
0
0
0
87
240
210
24
0
...
0



ANN connects every pixel to every neuron. 49*49 = 2401 49*10 = 490

To handle variety in digits we can use simple artificial neural network (ANN)

Variation in handwriting makes direct pixel matching unreliable.

Image size = 1920 x 1080 X 3

First layer neurons = 1920 x 1080 X 3 ~ 6 million

Hidden layer neurons = Let's say you keep it ~ 4 million

Weights between input and hidden layer = 6 mil * 4 mil
= 24 million

ANN treats all pixels equally.

ANN becomes inefficient when the number of parameters increases.

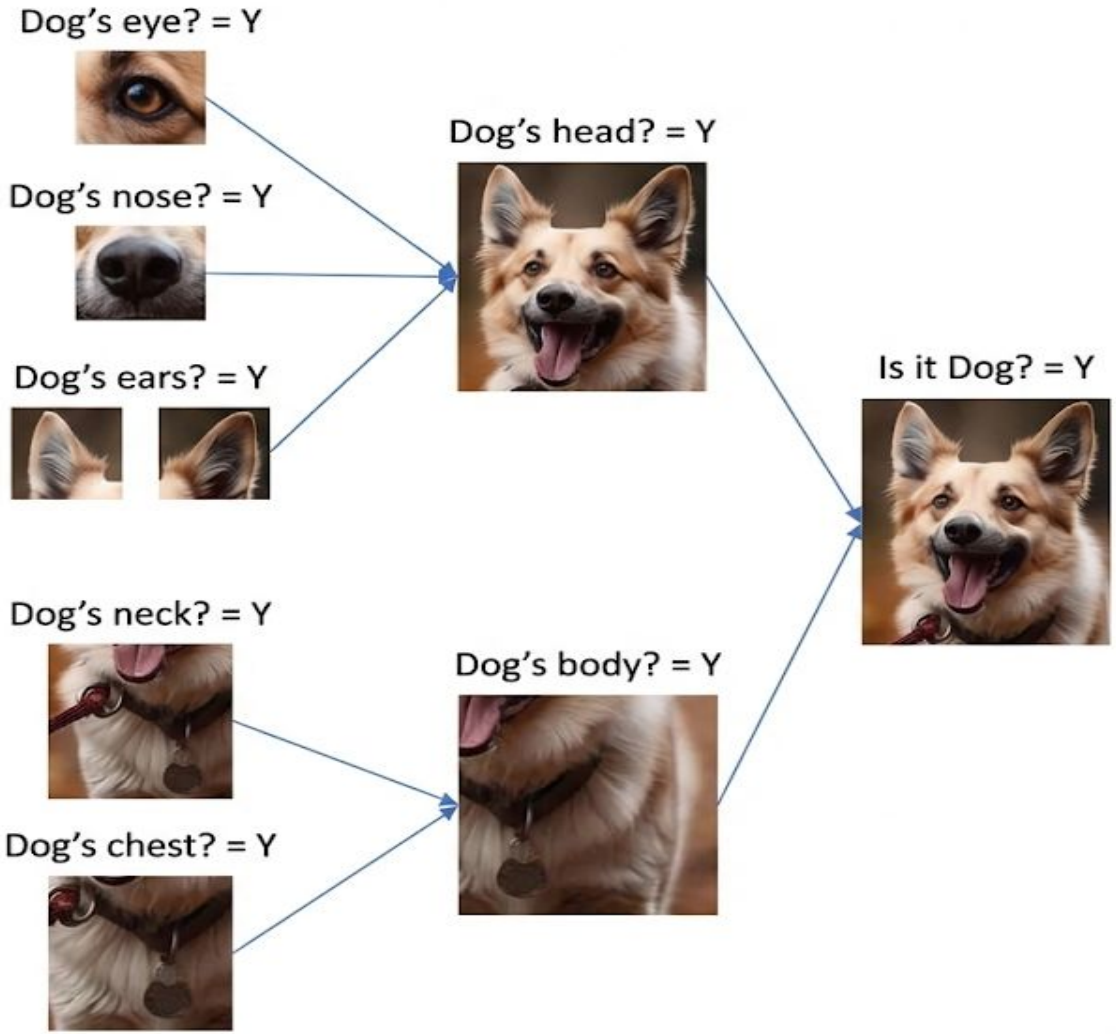
Disadvantages of using ANN for image classification

1. Too much computation
2. Treats local pixels same as pixels far apart
3. Sensitive to location of an object in an image

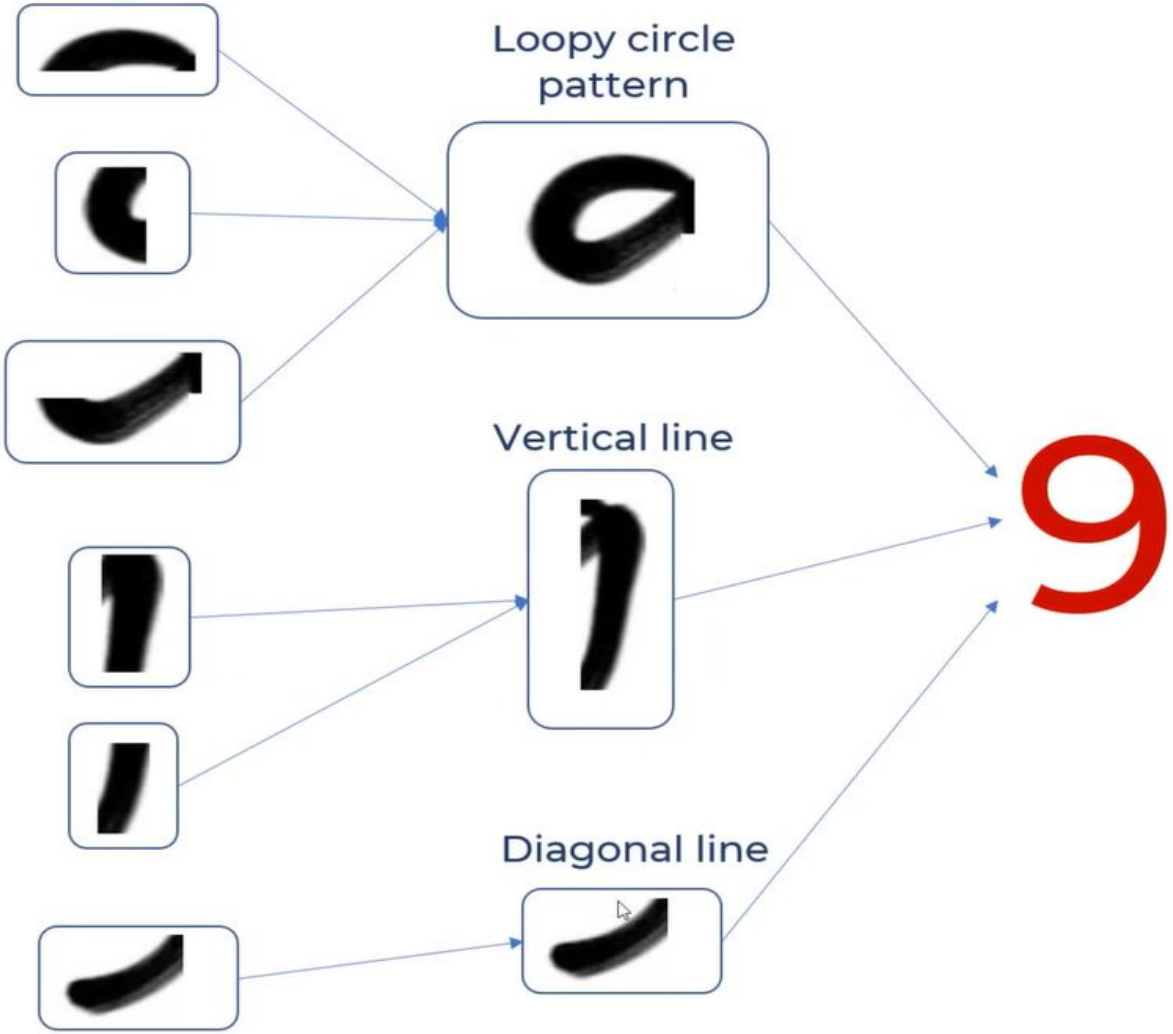
More pixels = more weights = more computation.

HOW DOES HUMANS
RECOGNIZE IMAGES SO EASILY?

SECTION 3: HUMAN VISION



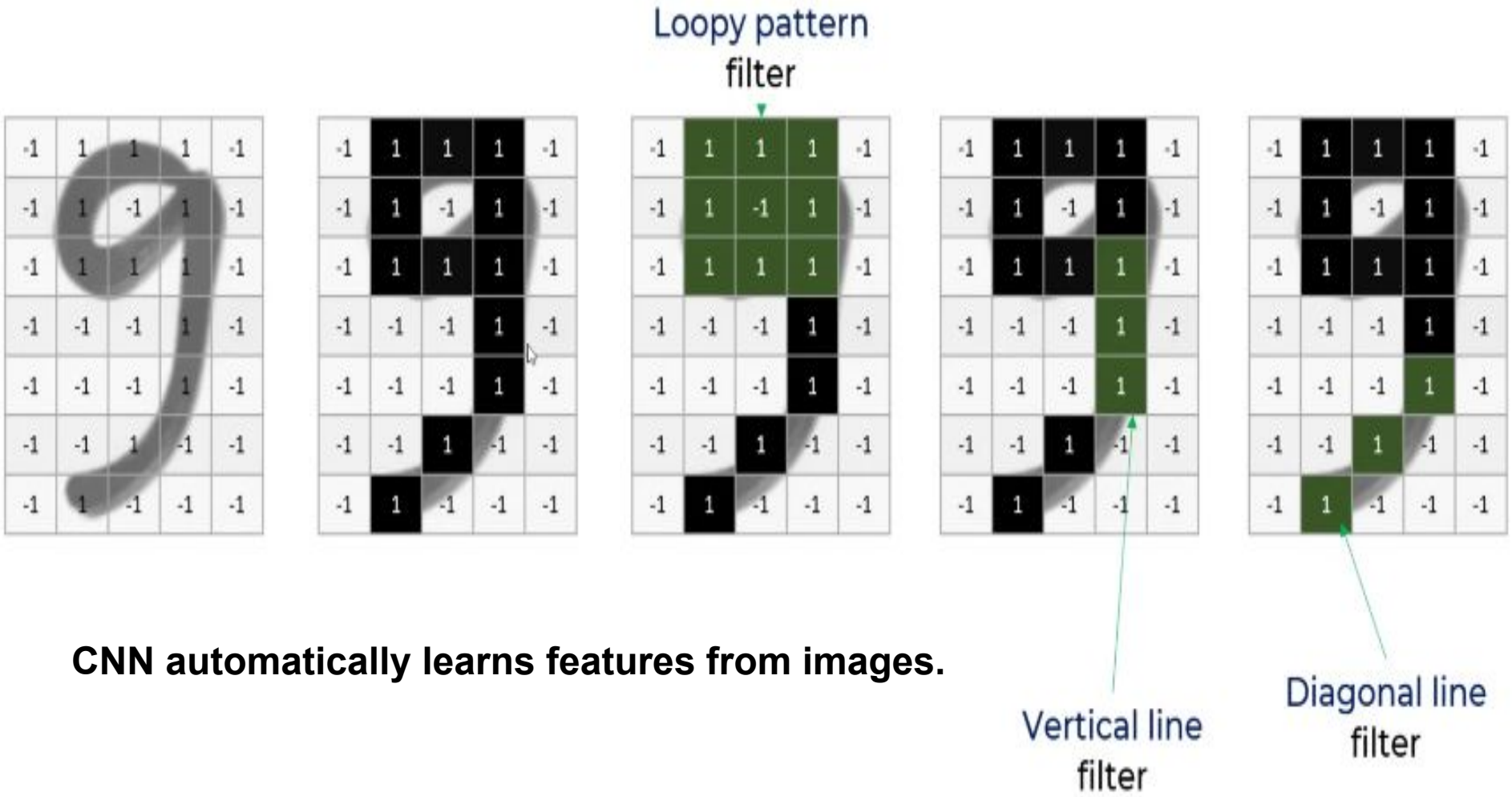
Humans focus on features, not pixels.



Objects are recognized by combining features

HOW CAN WE MAKE COMPUTERS
RECOGNIZE THESE TINY FEATURES?

SECTION 4: CONVOLUTION & FILTERS



CNN automatically learns features from images.

$$-1+1+1-1-1-1-1+1+1 = -1 \rightarrow -1/9 = -0.11$$

-1	**1**	**1**	**1**	-1
-1	**1**	-1	**1**	-1
-1	**1**	**1**	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	-1	**1**	-1
-1	-1	**1**	-1	-1
-1	**1**	-1	-1	-1

*

1	1	1
1	-1	1
1	1	1

-0.11		

CNN looks at small parts instead of whole image.

-1	1	1	1	-1
-1	1	-1	1	-1
-1	1	1	1	-1
-1	-1	-1	1	-1
-1	-1	-1	1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1

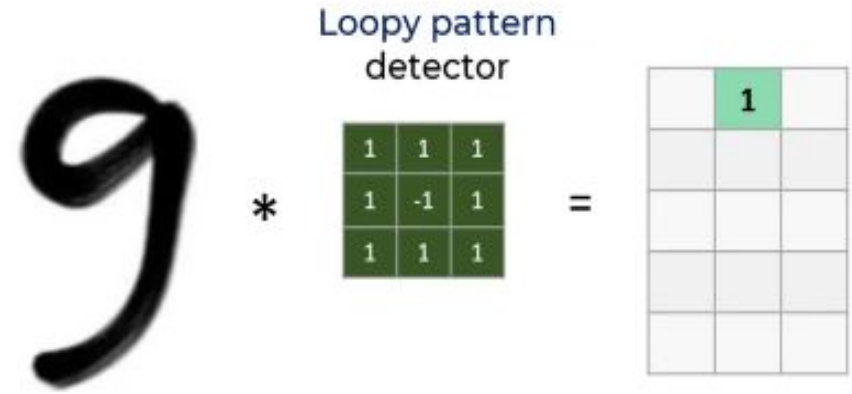
*

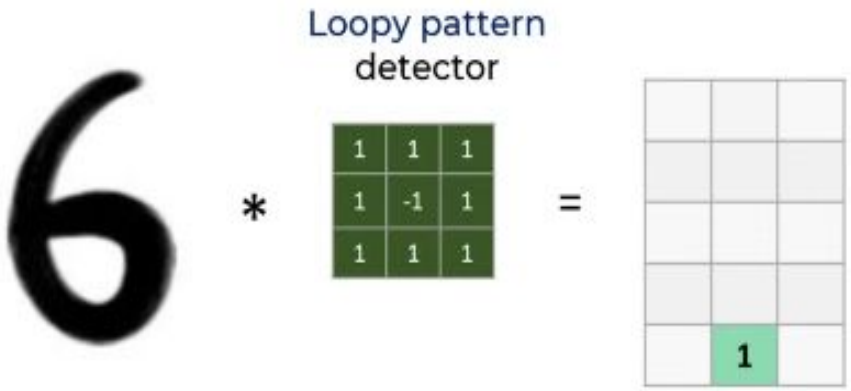
1	1	1
1	-1	1
1	1	1

-0.11	1	-0.11
-0.55	0.11	-0.33
-0.33	0.33	-0.33
-0.22	-0.11	-0.22
-0.33	-0.33	-0.33

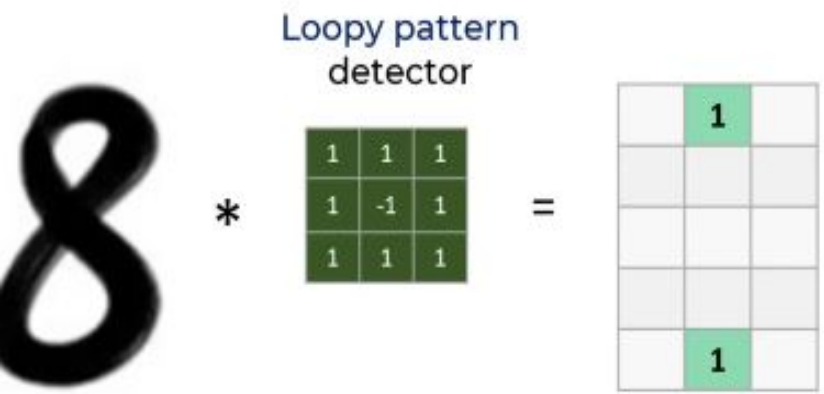
Feature Map

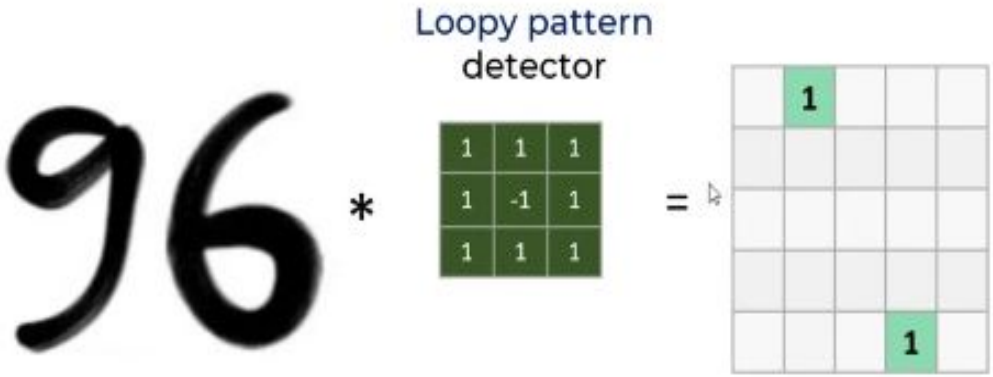
By focusing on small regions, CNN captures important local patterns in the image.





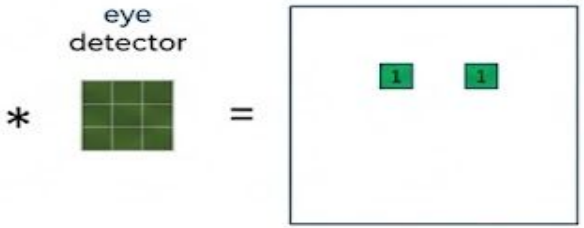
Filters are small matrices used to detect patterns.



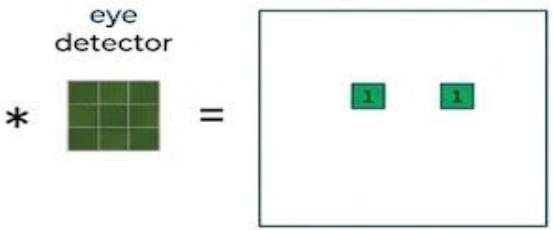


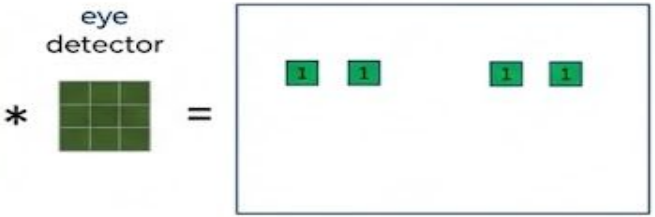
Filters are nothing but the feature detectors

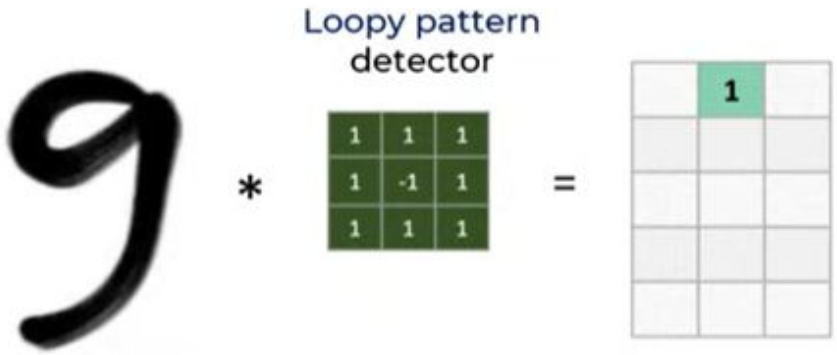
Location invariant: It can detect eyes in any location of the image

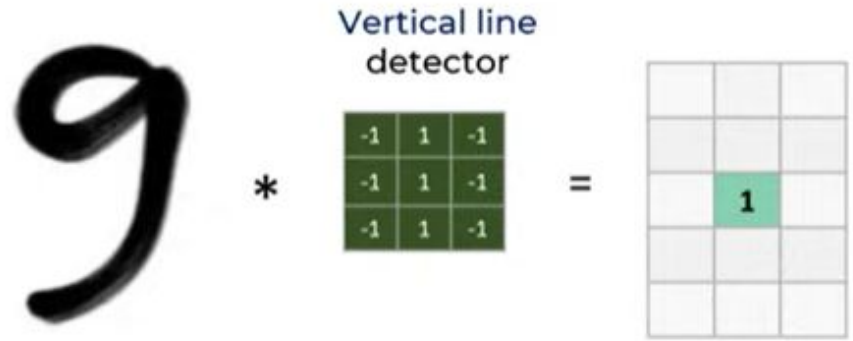


Dot product measures how strongly a filter matches a part of the image.

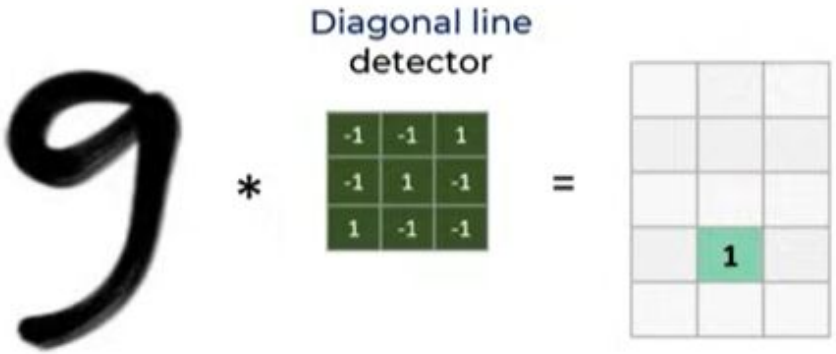




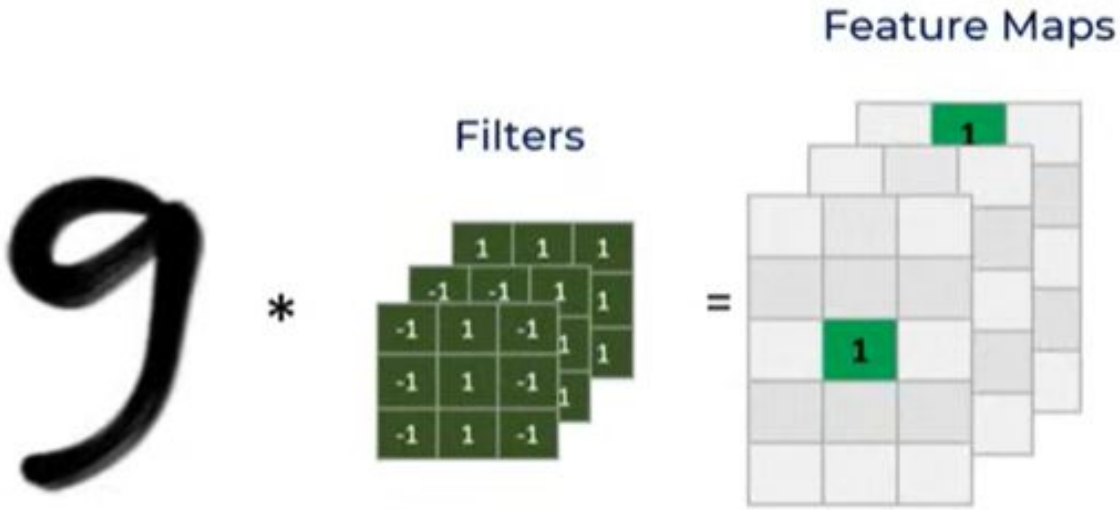


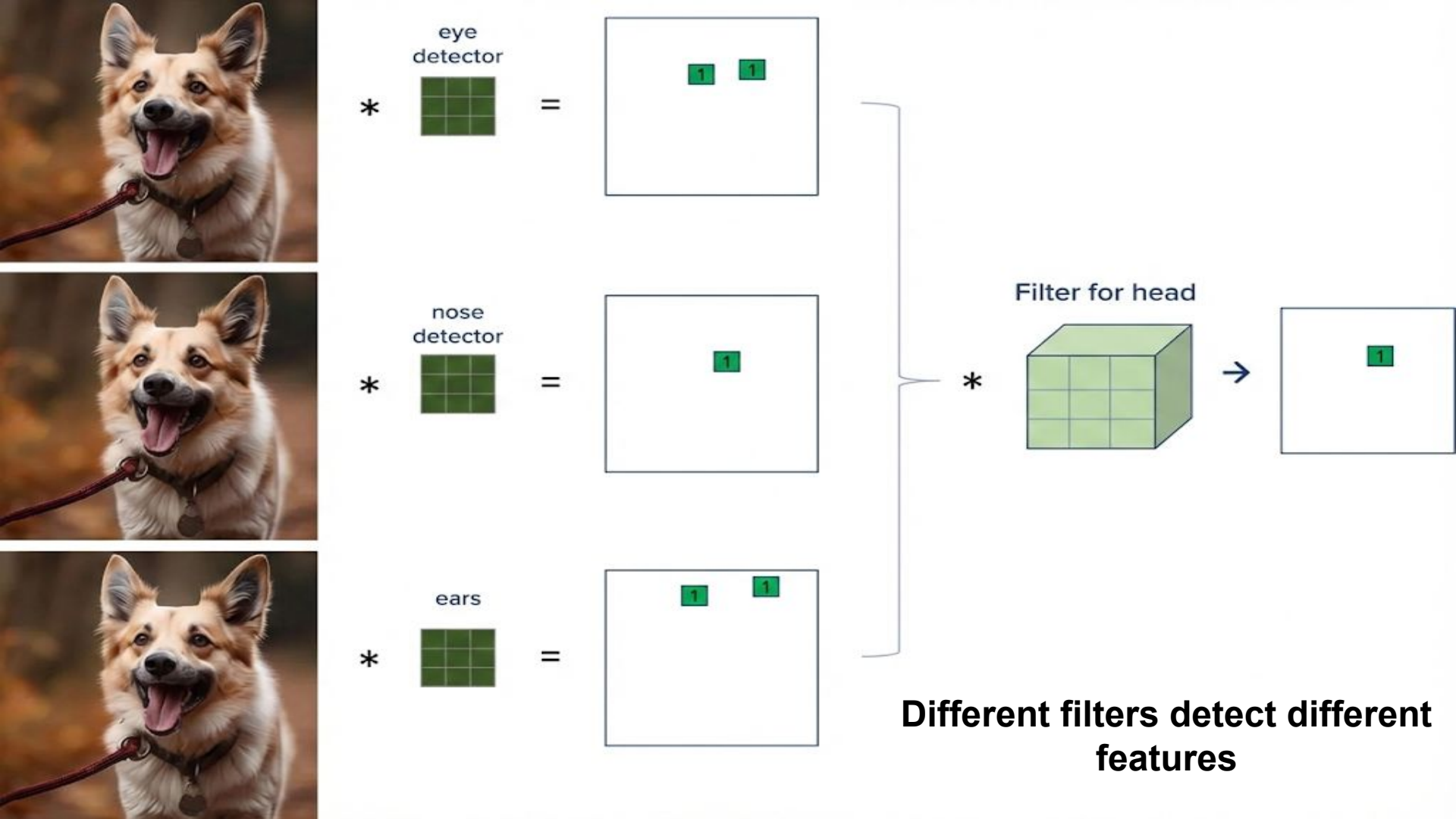


One value shows how strong the feature is.

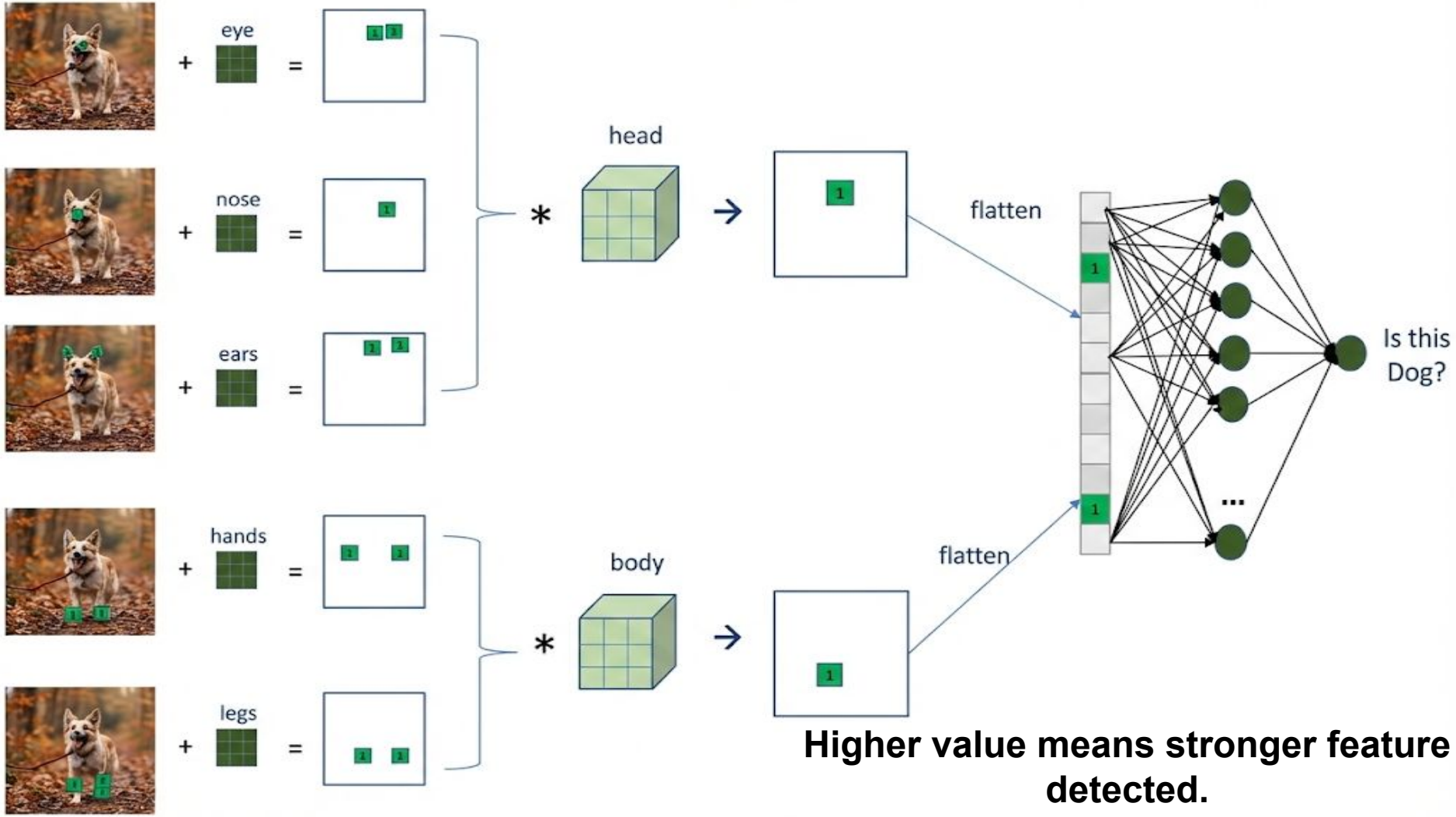


Repeating convolution creates a feature map.





Different filters detect different features



Higher value means stronger feature detected.

We didn’t address the issue of too much computation

SECTION 5: CNN PIPELINE

Pooling layer is used to reduce the size

5	1	3	4
8	2	9	2
1	3	0	1
2	2	2	0

8	**9**
3	2

2 by 2 filter with stride = 2

Shifted 9 at different position

1	1	1	-1	-1
1	-1	1	-1	-1
1	1	1	-1	-1
-1	-1	1	-1	-1
-1	-1	1	-1	-1
-1	1	-1	-1	-1
1	-1	-1	-1	-1

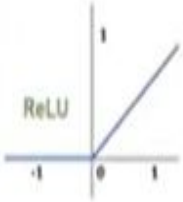
*

Loopy pattern filter

1	1	1
1	-1	1
1	1	1

→

1	-0.11	-0.11
0.11	-0.33	0.33
0.33	-0.33	-0.33
-0.11	-0.55	-0.33
-0.55	-0.33	-0.55



→

1	0	0
0.11	0	0.33
0.33	0	0
0	0	0
0	0	0

Max pooling

→

1	0.33
0.33	0.33
0.33	0
0	0

Final layers combine features to predict class

There is average pooling also...

5	1	3	4
8	2	9	2
1	3	0	1
2	2	2	0

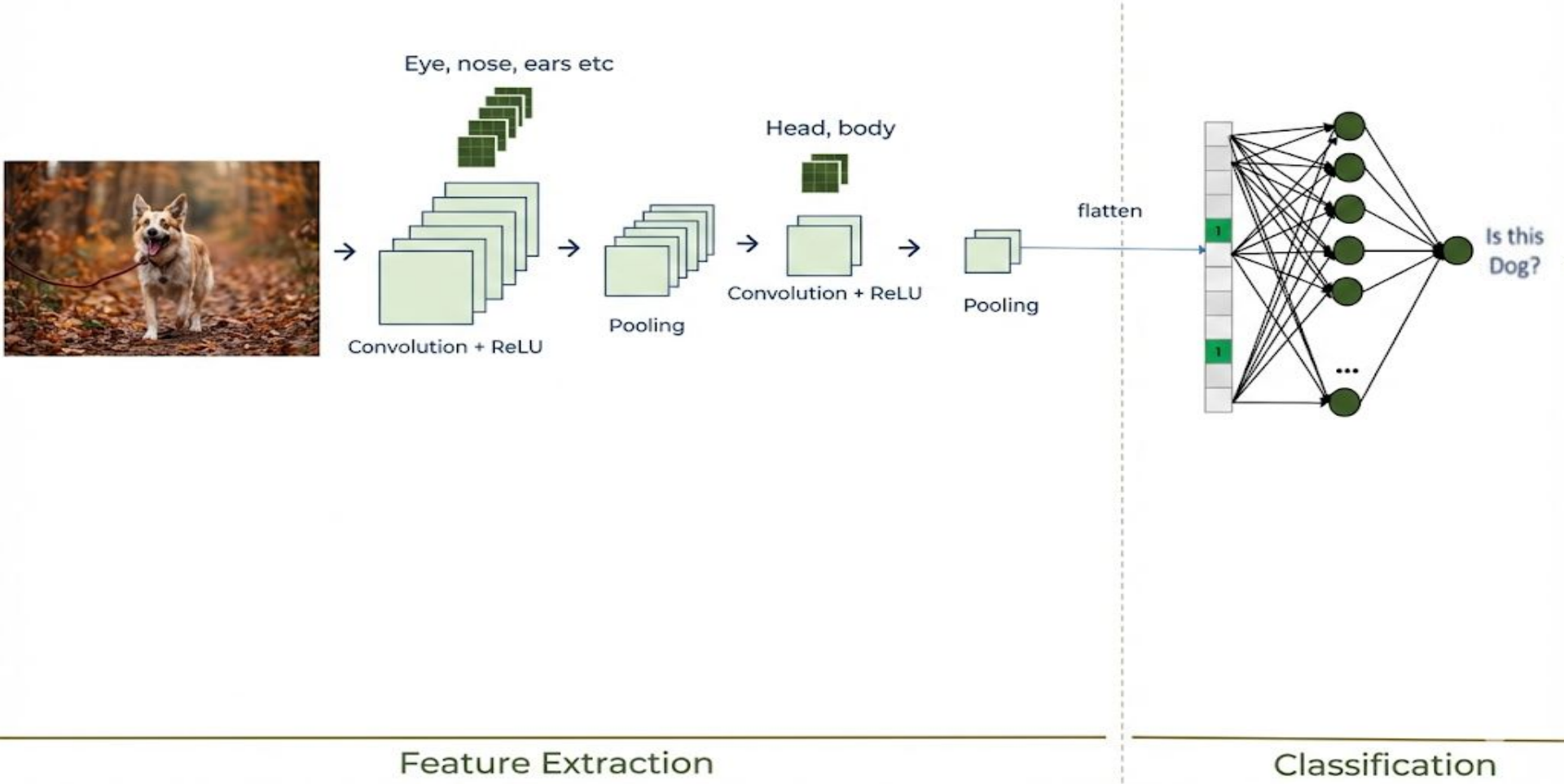
4	4.5
2	0.75

Benefits of pooling

Reduces dimensions & computation

Reduce overfitting as there are less parameters

Model is tolerant towards variations, distortions


Eye, nose, ears etc
Head, body
Convolution + ReLU
Pooling
Convolution + ReLU
Pooling
flatten
1
1
...
Is this Dog?
Feature Extraction
Classification